

JPE Replication Report

PUBLISHED
August 31, 2026

1 Paper Identification

ID	2016256
Author	Mitman
Title	Unemployment Benefits and Unemployment in the Great Recession: The Role of Macro Effects
Iteration	2

2 Summary

In assessing compliance with our [Data and Code Availability Policy](#), our reproducibility team has identified the following issues, which we ask you to address:

3 General

- ☒ Data Availability and Provenance Statements
 - ☒ Statement about Rights - Part 1: Right to use the data (e.g. *did the authors lawfully obtain the data*)
 - ☒ Statement about Rights - Part 2: Right to publish the data (e.g. *are human subjects involved and did permission been granted to publish their data?*)
 - ☒ License for Data (optional, but recommended)
 - ☒ Details on each Data Source
- ☒ Dataset list
- ☒ Computational requirements
 - ☒ Software Requirements
 - ☒ Controlled Randomness (as necessary)
 - ☒ Memory, Runtime, and Storage Requirements
- ☒ Description of programs/code
 - ☒ License for Code (Optional, but recommended)
- ☒ List of tables and programs
- ☒ References (Optional, but recommended)

4 High-level Description of Research Project

This research project...

- ☒ uses observational data.
- ☒ uses *confidential* observational data.
- ☐ uses data generated via experiment or trial by the authors (in field or lab).
 - ☐ For lab experiments: The code to implement the experimental protocol (z-Tree, o-Tree etc) is included in the package.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF document outlining the design of the experiment, including information on the selection and eligibility of participants is included.
 - ☐ A PDF copy of the instructions given to participants in both original language and an English translation is included.
- ☐ uses data generated via survey by the authors (in field or online)
 - ☐ The programs used to administer the survey (e.g. qualtrics survey file) are included.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF of the original questionnaire(s) and information on the selection and eligibility of participants is included.
- ☐ is a simulation study: the data are generated by an algorithm.
- ☐ is a numerical illustration: no data are used but code is used to produce illustrative output (tables, figures)

5 Data description

5.1 Data Sources

file/name	public	private	DOI/URL	Access described	cited readme	cited paper
BLS LAUS	yes	no	yes	yes	yes	yes
BLS JOLTS	yes	no	yes	yes	yes	yes
BLS QCEW	yes	no	yes	yes	yes	yes
Census LEHD QWI	yes	no	yes	yes	yes	yes
Census LODES	yes	no	yes	yes	yes	yes
BEA state GDP	yes	no	yes	yes	yes	yes
BEA CA25N industry employment	yes	no	yes	yes	yes	yes
FHFA HPI	yes	no	yes	yes	yes	yes
IRS SOI county income (via replication archive)	yes	no	yes	yes	yes	yes
NY Fed / Equifax county debt-to-income	yes	no	yes	yes	yes	yes
Saiz (2010) housing supply elasticity and WRLURI	yes	no	yes	yes	yes	yes
Census state tax revenue	yes	no	yes	yes	yes	yes
State business-policy indices	yes	no	yes	yes	yes	yes
USDA SNAP policy database	yes	no	yes	yes	yes	yes
Recovery.gov ARRA awards + geocorr crosswalk	yes	no	yes	yes	yes	yes
Census population estimates	yes	no	yes	yes	yes	yes
Shimer CPS separation series	yes	no	yes	yes	yes	yes
DOL / state UI benefit policy triggers	yes	no	yes	yes	yes	yes
County UI claims + UI tax receipts	yes	no	yes	yes	yes	yes
Census geography	yes	no	yes	yes	yes	yes
Conference Board HWOL vacancies (proprietary; synthetic stand-in provided)	no	yes	yes	yes	yes	yes

6 Requirements

- ✓ README is in TXT, MD, or PDF format
- ✓ README is accessible at the root of the package (not inside a folder)
- ✓ Title conforms to guidance (starts with “Data and Code for: Title of Paper” or “Code for: Title of Paper”, is properly capitalized)
- ✓ Authors (with affiliations) are listed in the same order as on the paper

6.1 Stated Requirements

- ☐ No requirements specified
- ✓ Software Requirements specified as follows:
 - **Stata/MP**, version 14 or later (the build and the `egen`-heavy steps assume MP; the seeded random draws use the mt64 generator, which is stable across Stata 14+). All required user-written commands install **automatically from SSC**: `code/config.do` — which every pipeline script sources — checks for and installs `regife` (interactive fixed effects), `reghdfe`, `ivreg2` (+ `ranktest`), `outreg2`, `binscatter`, `ftools`, `coefplot`, `tuples`, `hdfe`, and `mat2txt` on first run (an internet connection is needed once). No packages are bundled; `regife` is **not** redistributed — it loads from SSC like the rest. `config.do` also sets `scheme s2color` so regenerated figures match the published styling (Stata 18+ defaults to the newer `stcolor` scheme, which looks different).
 - **MATLAB** (developed and tested on R2025b) for the factor model, block bootstrap, and the structural model. Point estimates are deterministic and reproduce across MATLAB versions; see “Controlled Randomness” for a version caveat affecting bootstrap p-values/standard errors only.
 - **Python 3** for `code/tex/make_tables.py` (standard library only). The two map figures additionally need `geopandas` + `matplotlib` (any recent version; the driver skips the maps with a warning if `geopandas` is absent).
 - A **TeX** distribution (`pdflatex` + `bibtex`) to compile the paper.
- ✓ Computational Requirements specified as follows:
 - Storage: ≈10 GB free disk
 - Memory: 16 GB is comfortable for every step (the reproducibility check ran on a 64 GB machine; nothing approaches that bound)
- ✓ Time Requirements specified as follows:
 - Runtime: ≈10–15 hours sequentially on a modern multi-core machine, of which: the from-raw input rebuild ≈1.5 h; the 31 factor-model front-ends with 200-rep bootstraps ≈5–15 min each; the 200-pairing Scrambles runs ≈1–2 h; the Monte Carlo and the structural model most of the remainder. The Stata build itself takes minutes.
- ✓ Requirements are complete.

7 Analyzing Data and Code Files

7.1 Code description

- ✓ The replication package contains a “main” or “master” file(s) which calls all other auxiliary programs.

7.1.1 Summary of output-producing code

The package contains a master code in [replication-package/replication-package/code/RunBuild.do](#) that runs the raw-data build steps and the processed-data construction stages. The paper tables are assembled by [replication-package/replication-package/code/tex/make_tables.py](#), which reads CSV outputs generated by the Stata and Matlab analysis scripts and writes .tex fragments into the output/tables directory. The main figures are produced directly by [replication-package/replication-package/code/analysis/Binscatter_Figures.do](#), [replication-package/replication-package/code/analysis/MO_Tightness_Figures.do](#), [replication-package/replication-package/code/analysis/Make_CountyMap.py](#), and [replication-package/replication-package/code/analysis/Make_BenefitMaps.py](#). Per the README’s exhibit classification, `Binscatter_Figures.do` (Figure 1), `Make_CountyMap.py` (Figure A-5), and `Make_BenefitMaps.py` (Figure A-6) reproduce exactly, while `MO_Tightness_Figures.do` (Figures A-1 and A-2, and the vacancy/tightness columns of Table A-1) depends on the proprietary Conference Board HWOL vacancy series, which is excluded from the archive under the authors’ data-use agreement; as shipped it runs on a public-data synthetic stand-in and so produces demonstrative, not published, output. A small set of inline policy numbers is derived in [replication-package/replication-package/code/analysis/Derived_Policy_Calcs.do](#), which reproduces exactly except that its tightness-scenario row re-uses the published HWOL-based coefficient (0.101) as a stated constant rather than re-estimating it.

7.1.2 Data Preparation Code

Program	Dataset or analysis file created	Reproduced?
replication-package/replication-package/code/RunBuild.do	Main orchestrator for the full raw-data rebuild and the processed-data build.	Yes

Program	Dataset or analysis file created	Reproduced?
replication-package/replication-package/code/build/MakeDataSetsMainPaper_v5_newvac.do	Builds the main analysis panel used in the paper regressions.	Yes
replication-package/replication-package/code/build/MakeDataControls.do	Builds the control-variable panel used by the analysis scripts.	Yes
replication-package/replication-package/code/analysis/Derived_Policy_Calcs.do	Writes the derived policy-scenario CSV that underlies the paper's inline policy numbers.	Yes

7.1.3 Tables and Figures

Output in paper	Program and line number(s)	Reproduced?
Table 1 / tab:Benefits_on_unemp	replication-package/replication-package/code/tex/make_tables.py	Yes
Table 2 / tab:Forward_Spec	replication-package/replication-package/code/tex/make_tables.py and replication-package/replication-package/code/matlab/ProcessQDK.m	Yes
Table 3 / tab:Benefits_on_JobCreation	replication-package/replication-package/code/tex/make_tables.py	Partial — col 3 (Employment) only; cols 1-2 (Vacancies, Tightness) require licensed HWOL data and are demonstrative as shipped
Table 4 / tab:Benefits_on_Wages	replication-package/replication-package/code/tex/make_tables.py	Yes
Figure 1(a) and Figure 1(b) / fig:binscatter_du and fig:binscatter_qdu	replication-package/replication-package/code/analysis/Binscatter_Figures.do	Yes
Figure A-1 / fig:MO_vacPuD	replication-package/replication-package/code/analysis/MO_Tightness_Figures.do	No — requires licensed HWOL state vacancy data; demonstrative as shipped
Figure A-2 / fig:Tightness_MO_Borders	replication-package/replication-package/code/analysis/MO_Tightness_Figures.do	No — requires licensed HWOL state vacancy data; demonstrative as shipped
Table A-1 / tab:MO_V_U	replication-package/replication-package/code/tex/make_tables.py and replication-package/replication-package/code/analysis/MO_Tightness_Figures.do	Partial — Unemployment column only; Vacancy and Tightness columns require licensed HWOL data
Figure A-3 / fig:initial-guess	replication-package/replication-package/code/matlab/Fig_InitialGuess.m	Yes
Table A-2 / tab:Monte-Carlo-Results	replication-package/replication-package/code/tex/make_tables.py and replication-package/replication-package/code/matlab/RunMonteCarlo.m	Yes
Table A-3 / tab:Benefits_on_unemp_OLS	replication-package/replication-package/code/tex/make_tables.py	Yes
Table A-4 / tab:Endog_Vars	replication-package/replication-package/code/tex/make_tables.py and replication-package/replication-package/code/analysis/Endog_Vars.do	Yes

Output in paper	Program and line number(s)	Reproduced?
Figure A-4 / fig:endogeneity_test_unemp	replication-package/replication-package/code/model/RunModel.m and replication-package/replication-package/code/model/Fig_StateU.m	Yes
Table A-7 / tab:Mobility_LODES	replication-package/replication-package/code/tex/make_tables.py and replication-package/replication-package/code/analysis/Mobility_LODE_S.do	Yes
/ tab:Imputed_Results	replication-package/replication-package/code/tex/make_tables.py and replication-package/replication-package/code/analysis/Imputed_Results.do	No — all columns run on the vacancy file (<code>\${vacfile}</code>); requires licensed HWOL data. The generated <code>.tex</code> is a comparison file only; the published table is typeset directly in the paper source
Table A-9 / app_tab:A-1	replication-package/replication-package/code/tex/make_tables.py and replication-package/replication-package/code/analysis/TableA2.do	Yes
Table A-11 / tab:Benefits_on_unemp_beg	replication-package/replication-package/code/tex/make_tables.py	Yes
Table A-13 / tab:Benefits_on_unemp_SNAP_Mortgage	replication-package/replication-package/code/tex/make_tables.py	Yes
Table A-14 / tab:Benefits_on_unemp_stimulus_taxes	replication-package/replication-package/code/tex/make_tables.py	Yes
Table A-15 / tab:Benefits_on_unemp_other_policies	replication-package/replication-package/code/tex/make_tables.py	Yes
Figure A-5 / fig:Counties_Map	replication-package/replication-package/code/analysis/Make_County_Map.py	Yes
Figure A-6 and Figure 6(a-j) / fig:Benefit_Maps and fig:Benefit_Maps_42 through fig:Benefit_Maps_96	replication-package/replication-package/code/analysis/Make_Benefit_Maps.py	Yes
Table A-16 / tab:app_Benefits_on_Wages	replication-package/replication-package/code/tex/make_tables.py	Yes

7.1.4 In-Text Numbers

“Narrative” means the number is an institutional fact, a cited value, or a hand calculation stated in the paper text; no program produces it, and none is expected to. Everything else was checked against the authors’ shipped output in `output/`.

In-text numbers	Program	Line Number	Reproduced?
16,000 (online job boards covered by HWOL)	Narrative — Conference Board HWOL documentation, no program	n/a	n/a — provider documentation, not computed. HWOL is confidential and not shipped
93% (HWOL vacancies matched to a county)	Narrative — Conference Board HWOL documentation, no program	n/a	n/a — provider documentation, not computed

In-text numbers	Program	Line Number	Reproduced?
5% (HWOL vacancies coded "statewide")	Narrative — Conference Board HWOL documentation, no program	n/a	n/a — provider documentation, not computed
2% (HWOL vacancies coded "nationwide")	Narrative — Conference Board HWOL documentation, no program	n/a	n/a — provider documentation, not computed
1,172 (border county pairs)	replication-package/replication-package/code/matlab/Factor_FrontEnd_Bench.m	L18	Yes — output/factor_results/Bench_qd1_unemp_Nobs.csv = 37,496 = 1,172 x 32 quarters
1,132 (pairs with different benefits in at least one quarter)	No program — recomputed by the replicator from data/processed/UIDMacro_DataControls.dta (diff_logmeanwks , 2008-2012)	n/a	No — the data give 1,142, not 1,132 (10 pairs more). Stable across tolerances from 1e-9 to 0.02, so it is not a rounding artifact
14 (median quarters with different durations)	No program — recomputed by the replicator from data/processed/UIDMacro_DataControls.dta	n/a	Yes — median = 14 quarters (2008-2012)
0 to 18 (range of quarters with different durations)	No program — recomputed by the replicator from data/processed/UIDMacro_DataControls.dta	n/a	Yes — range = 0 to 18 quarters
1,113 (counties below 15% of state employment)	replication-package/replication-package/code/analysis/Bias_Sim_Shares.do	L29	No — the script returns 1,108 (output/factor_results/Bias_Sim_Shares.csv). The companion 2% average share does reproduce (2.0%)
59 (counties at or above 15% of state employment)	replication-package/replication-package/code/analysis/Bias_Sim_Shares.do	L34	No — the script returns 64 (output/factor_results/Bias_Sim_Shares.csv). The companion 35% average share comes out at 34.3%. Note 1,108 + 64 = 1,172, so the split moves but the total is right
282 / 9,024 (single-county-LMA claims sample)	replication-package/replication-package/code/exporters/OutputDataSetsUIDMacro_JFR.do	L21-L26	Yes — output/factor_results/JFR_jfr_unemp_Nobs.csv = 9,024 = 282 x 32
26 to 99 (weeks, maximum benefit duration)	Narrative — UI programme description, no program	n/a	n/a — policy fact
13 or 20 (extra weeks under EB)	Narrative — UI programme description, no program	n/a	n/a — policy fact
4 (EUC08 tiers)	Narrative — UI programme description, no program	n/a	n/a — policy fact
10.7% (sequester cut)	Narrative — 2013 budget sequestration, no program	n/a	n/a — policy fact; the paper states it but does not use it

In-text numbers	Program	Line Number	Reproduced?
to EB/EUC08 spending)			
16 (weeks cut in Missouri, April 2011)	Narrative — cited from Johnston and Mas (2015), no program	n/a	n/a — cited value; it is the event date marked in Figures A-1/A-2
13 (initial extra weeks under EUC08; also the placebo extension)	replication-package/replication-package/code/build/Make_PlaceboWeeksData.do	L9, L43	Yes as used — the placebo build sets 26 + 13 = 39 weeks exactly as the text describes
3.02 and 2.15 (pp lower unemployment in 2010/2011)	replication-package/replication-package/code/analysis/Derived_Policy_Calcs.do	L109-L113	Yes — 3.0186 and 2.1545 (output/factor_results/Derived_Policy_Calcs.csv)
5% to 9.6% (long-run unemployment, permanent 26 to 99)	replication-package/replication-package/code/analysis/Derived_Policy_Calcs.do	L48, L53-L55	Yes — 9.6023 (Derived_Policy_Calcs.csv); the 5% base is an assumption set at L48
95% to 90.4% (long-run employment analogue)	replication-package/replication-package/code/analysis/Derived_Policy_Calcs.do	L56-L57	Yes — 90.3977 (Derived_Policy_Calcs.csv)
10% (average quarterly JOLTS separation rate)	replication-package/replication-package/code/analysis/Derived_Policy_Calcs.do	L47	Yes — hard-coded as 0.10; the replicator checked it against the shipped data/raw/jolts2013.dta , whose mean quarterly separation rate is 0.1048
0.4 to 0.5 (consensus matching-function elasticity)	Narrative — cited from Brügemann (2008) and Petrongolo and Pissarides (2001), no program	n/a	n/a — cited range; the midpoint 0.45 is used at Derived_Policy_Calcs.do L61
0.0528 (unemployment change via the tightness margin)	replication-package/replication-package/code/analysis/Derived_Policy_Calcs.do	L60-L62	Yes — 0.05277. Caveat: the tightness coefficient 0.101 is entered at L60 as a published constant, because re-estimating it needs the confidential HWOL data
73 (weeks of extension, 99 minus 26)	Narrative — arithmetic in the text, no program	n/a	n/a — arithmetic
7.3 to 14.6 (implied extra weeks of unemployment duration)	Narrative — 73 x 0.1 and 73 x 0.2, arithmetic in the text	n/a	n/a — arithmetic on elasticities taken from the micro literature review
5/27 (zeta, non-participant search margin)	replication-package/replication-package/code/analysis/Impute_Input.do and impute_border_search.m	Impute_Input.do L104; impute_border_search.m L60	Yes as used — calibrated constant from Hall (2013), coded as 5/27 in both places

In-text numbers	Program	Line Number	Reproduced?
0.78% (wage gap, 70 vs 50 weeks of benefits)	replication-package/replication-package/code/matlab/Factor_FrontEnd_QWI_WStayers.m via make_tables.py	make_tables.py L737-L827	Yes — the input coefficient is Table A-16 Column (1), which reproduces at 0.0232; $\exp(0.0232 \times (\ln 70 - \ln 50)) = 1.0078$, i.e. 0.78%. The final step is arithmetic in the footnote, not in code
0.101 (tightness response to benefit duration, used in the 0.0528 calculation)	replication-package/replication-package/code/analysis/Derived_Policy_Calcs.do — published source is Table 3 Column (2), make_tables.py	L60	No — cannot be checked. Table 3 Column (2) needs the confidential HWOL vacancy data, so the coefficient is typed into the script as a constant rather than re-estimated. On the shipped synthetic stand-in the same column comes out at -0.089, which is demonstrative only
0.951 and 0.316 (Hall endogeneity test, LAUS unemployment rate on state and adjacent county)	replication-package/replication-package/code/analysis/TableA2.do	L110-L168	Yes — 0.9507 and 0.3161 (output/factor_results/Hall_Inline.csv , row laus_rate)
1.121 and 0.415 (same test on continuing claims divided by population)	replication-package/replication-package/code/analysis/TableA2.do	L113-L160	No — the script returns 0.891 and 0.312 (output/factor_results/Hall_Inline.csv , row claims_pop). Not a data-access problem: the claims file ships (README dataset list, row 19). The script header at L114-L121 states the original program was lost and that this is a reconstruction using an all-county state aggregate, so the published magnitudes cannot be recovered until the authors supply the original construction. The paper's conclusion — claims show the same large-state pattern as LAUS — still holds
30 (miles, median distance between population centres)	replication-package/replication-package/code/exporters/OutputDataSetsUIMacro_Dist30.do	L24	Yes — the replicator computed the median of dist in UIMacro_DataControls.dta at 29.62 miles, which rounds to the stated 30
10,000 (minimum core population of a CBSA)	Narrative — OMB (2010) CBSA definition, no program	n/a	n/a — definitional

Summary: the table lists 32 numbers. 13 are narrative (policy facts, cited values, or arithmetic stated in the text) and need no code. Of the 19 that are data- or code-based, 14 reproduce, 4 do not, and 1 cannot be checked at all.

The four misses are **1,132** (comes out at 1,142), **1,113** (comes out at 1,108), **59** (comes out at 64), and the Hall claims pair **1.121 / 0.415** (comes out at 0.891 / 0.312). The one number that cannot be checked is the tightness coefficient **0.101** behind the 0.0528 calculation. All four misses can be re-derived from the shipped data with [verify/recompute_intext.do](#), which the replicator ran in Stata/MP 19.5 with no errors; a Python equivalent, [verify/recompute_intext.py](#), returns the same figures. The Stata run matches the authors' own shipped [Hall_Inline.csv](#) to seven decimals (0.8911967 and 0.3116272, N=9,344), confirming the mismatch is in the reconstruction itself and not in the replicator's re-implementation.

Only the tightness coefficient is caused by the withheld HWOL data. The other four are not, and should not be reported as licensing casualties:

- **1,132**: output from [diff_logmeanwks](#), built from [benefit_weeks.dta](#), which ships and which the README states is identical to the authors' own ICPSR deposit. The gap is 10 pairs that differ in only a few quarters out of 20; 23 pairs differ in 3 quarters or fewer, so a small shift in the benefit-weeks series moves the count. It is not due to rounding.

- **1,113 / 59**: output from `emp_share`, defined at [MakeDataSetsMainPaper_v5_newvac.do L164-L166](#) as `emp_laue/emp_state_u` in 2012q4 — LAUS county employment over BLS state employment, both shipped and both vintage-stamped. The gap is at the 0.15 cutoff and runs upward: the archived vintage places five *more* pairs at or above 0.15 than the published text did. Recovering 1,113 / 59 requires the five lowest pairs now above the line — 0.1502, 0.1507, 0.1607, 0.1607 and 0.1615 — to fall below it. Two are within 0.001 of the cutoff, but three sit about 0.011 above it, so this is not pure rounding: the employment ratio for those three would need to be roughly 7% lower in the authors' vintage.
- **1.121 / 0.415**: the continuing-claims file ships (README dataset list, row 19). The authors' own script header records that the original producing program was lost and that the shipped version is a reconstruction using an all-county state aggregate.

The synthetic vacancy stand-in is merged into the panel at [MakeDataSetsMainPaper_v5_newvac.do L133](#), but that merge only adds vacancy columns and drops no rows, so it cannot move any of these four counts. All four are deterministic given the shipped inputs: re-running the package will reproduce 1,142 / 1,108 / 64 / 0.891 / 0.312 every time. Note also that $1,108 + 64 = 1,172$ and $1,142 + 30 = 1,172$, so the pair sample itself is intact; only the underlying values shifted. The likeliest explanation for the first three is that the published text was written against an earlier LAUS and benefit-weeks vintage than the one archived here. The authors should either update the text to the archived vintage or state which vintage the published numbers came from.

7.2 Computing Environment of the Replicator

- Macbook Air 2020 M1, 8GB RAM
- Stata/MP 19.5
- Matlab R2026a
- Python 3.11.5
- pdfTeX 3.141592653-2.6-1.40.27

8 Replication

The code and data were accessed via the provided replication package. The code ran smoothly without any modifications needed.

8.1 Findings

All tables and figures were reproduced except: Table 3 columns 1–2 (Vacancies, Tightness), Table A-1 (Vacancy and Tightness columns), Table A-6 (Data-row Vacancy/Tightness cells), Table A-8 (all columns), Table A-12 columns 1–2, and Figures A-1 and A-2, as stated by the authors. Styling issues persist for the descriptive figures A-3, A-5, A-6, as acknowledged by the authors.

8.2 Data Preparation Code

Programs ran without errors.

8.3 Tables and Figures

8.3.1 Figures

Figure A-3 ([initial_guess.pdf](#))

Figure A-3: Irrelevance of the initial guess for the factor model estimation

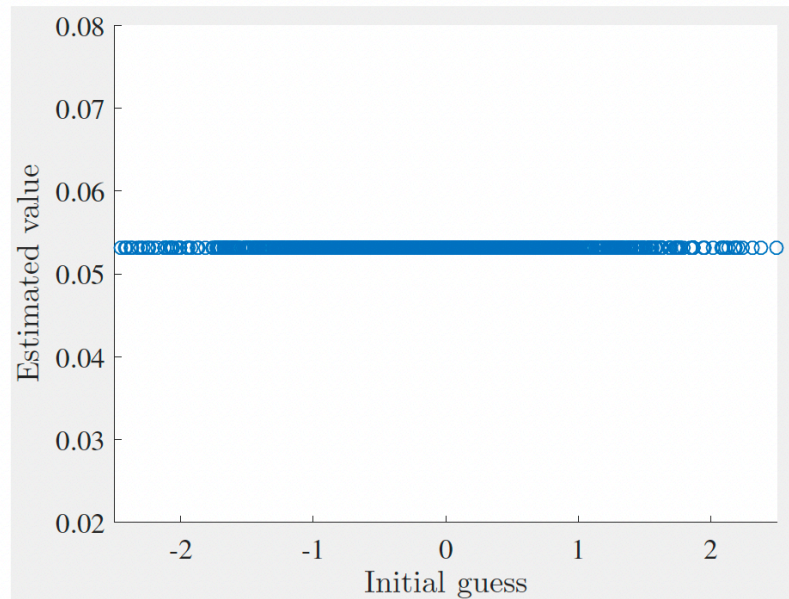
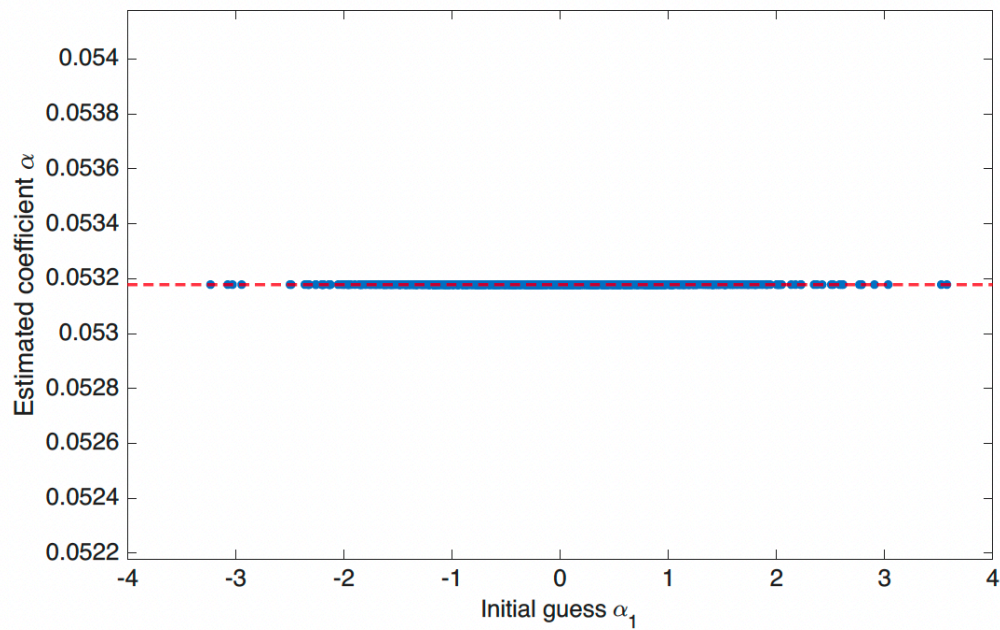
*paper**Replication*

Figure 1: Figure A-3

Finding: The reproduced version differs a lot in style but also in naming of the axes.

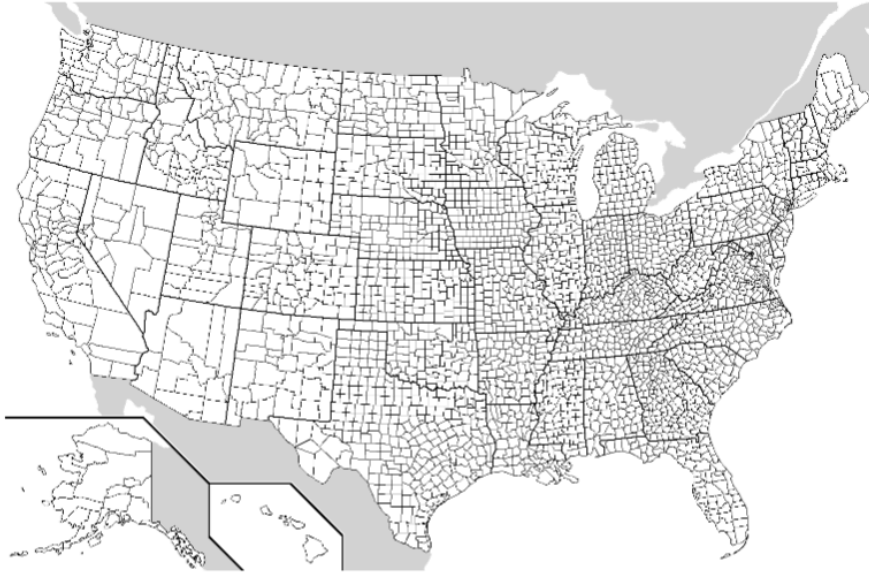


Figure A-5: Map of U.S.A. with state and county outlines.

paper

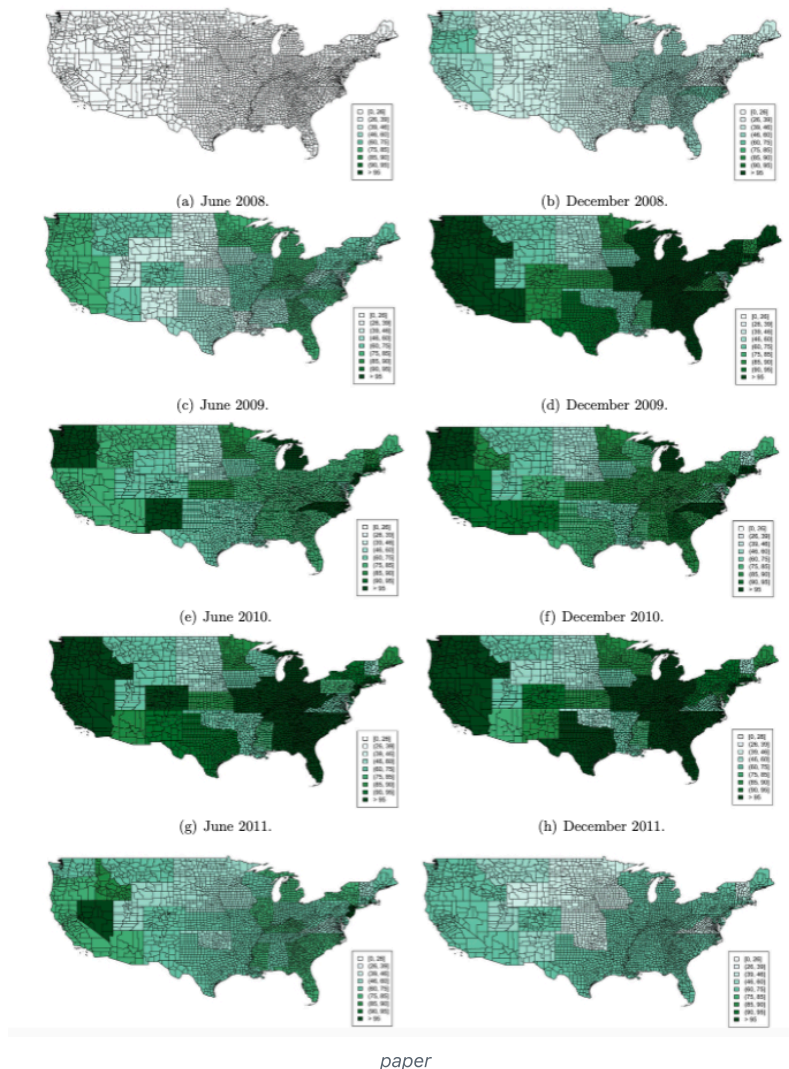


Figure A-5: Map of U.S.A. with state and county outlines.

Replication

Figure 2: Figure A-5

Finding: Reproduced, but features a completely different style and look compared to the published paper.



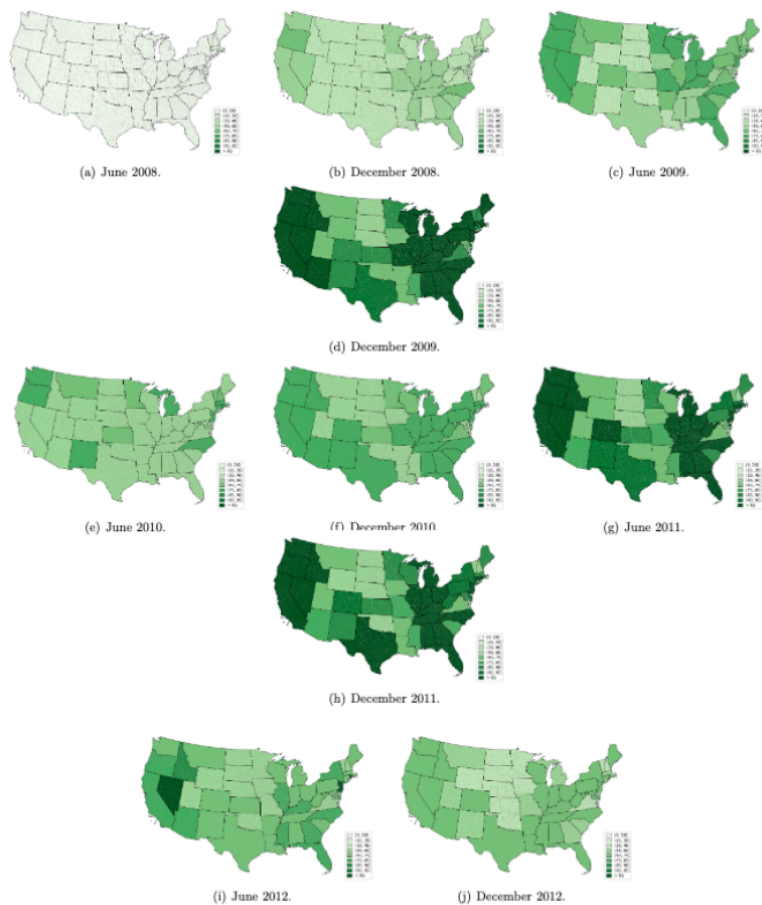


Figure A-6: Unemployment benefit duration across U.S. states during the Great Recession. Selected months.

Replication

Figure 3: Figure A-6

Finding: Reproduced, but features a completely different style and look compared to the published paper.

8.4 Other issues

- Three in-text numbers couldn't be reproduced - please check the issue in the previous section.

9 Classification

Full reproduction can include a small number of apparently insignificant changes in the numbers in the table. Full reproduction also applies when changes to the programs needed to be made, but were successfully implemented.

Partial reproduction means that a significant number (>25%) of programs and/or numbers are different.

Note that if some data is confidential and not available, then a partial reproduction applies. This should be noted in the Reasons.

Note that when all data is confidential, it is unlikely that this exercise should have been attempted.

Failure to reproduce: only a small number of programs ran successfully, or only a small number of numbers were successfully generated (<25%). This also applies when all data is restricted-access and none of the **main** tables/figures are run.

- ☐ full reproduction
- ☐ full reproduction with minor issues
- ☒ partial reproduction (see above)
- ☐ not able to reproduce most or all of the results (reasons see above)

9.1 Reason for incomplete reproducibility

- ☐ None.
- ☒ **Discrepancy in output** (either figures or numbers in tables or text differ)
- ☐ **Bugs in code** that were fixable by the replicator (but should be fixed in the final deposit)
- ☐ **Code missing**, in particular if it prevented the replicator from completing the reproducibility check
 - ☐ **Data preparation code missing** should be checked if the code missing seems to be data preparation code
- ☐ **Code not functional** is more severe than a simple bug: it prevented the replicator from completing the reproducibility check
- ☐ **Software not available to replicator** may happen for a variety of reasons, but in particular (a) when the software is commercial, and the replicator does not have access to a licensed copy, or (b) the software is open-source, but a specific version required to conduct the reproducibility check is not available.
- ☐ **Insufficient time available to replicator** is applicable when (a) running the code would take weeks or more (b) running the code might take less time if sufficient compute resources were to be brought to bear, but no such resources can be accessed in a timely fashion (c) the replication package is very complex, and following all (manual and scripted) steps would take too long.
- ☐ **Data missing** is marked when data *should* be available, but was erroneously not provided, or is not accessible via the procedures described in the replication package
- ☒ **Data not available** is marked when data requires additional access steps, for instance purchase or application procedure.
- ☐ **Missing README** is marked if there is no README to guide the replicator, or the README is not in compliance with JPE requirements